

BST Physics Curriculum Vol 10 Chapter 10 — Calculus of Variations + Path Integrals v0.4 (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 10 primary]

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Vol 10 Chapter 10 — Calculus of Variations + Path Integrals

Chapter motivation

Standard graduate calculus of variations: stationary-action principle $\delta S = 0$; Euler-Lagrange equations; Hamilton's principle; Noether's theorem (symmetries $\rightarrow$ conservation laws). Feynman path integral (1948): quantum amplitudes as $\int Dq(t) \exp(iS[q]/\hbar)$ sum over all paths; Wick rotation $t \rightarrow -i\tau$ to convergent Gaussian. Standard texts: Gelfand-Fomin + Goldstein + Feynman-Hibbs + Kleinert.

BST cross-link: per Vol 5 Ch 5 + Vol 1 Ch 7 + Vol 1 Ch 9 + T2457 (Friday): Feynman path integral becomes finite sum over substrate ticks via T2429 $GF(128)^k$ discretization; Bergman reproducing kernel $K_B(z, \bar{w})$ plays structural role of QFT Feynman propagator per T2457 Cal #92(b) framing (positive-definite + UV-complete + BST primary normalization $c_{FK} = 225/\pi^{(9/2)}$); no Wick rotation needed (substrate-tick is real-time finite-step per T2405 Koons tick $\sim 10^{-120}$ s).

Section 10.0b — Reader-grade 3-level pedagogy

Level 1: Standard variational principles (Euler-Lagrange + Hamilton's principle + Noether) + Feynman path integral; BST cross-link Vol 5 Ch 5: substrate path integral = finite sum over substrate-tick $GF(128)^k$ operations; Bergman kernel plays propagator role per T2457; no Wick rotation needed.

Level 2 (graduate-physicist): Standard variational principles: stationary-action $\delta S = 0$ with $S = \int L dt$; Euler-Lagrange $\partial L / \partial q - d/dt (\partial L / \partial \dot{q}) = 0$; Hamilton's principle $\delta \int (T - V) dt = 0$; Lagrangian $= T - V$ (kinetic minus potential); Legendre transform Hamiltonian $H = p\dot{q} - L$. Noether's theorem: continuous symmetry $\rightarrow$ conserved charge; e.g., time-translation symmetry $\rightarrow$ energy conservation (T2473 substrate-derivation Friday). Feynman path integral (1948 Nobel 1965): quantum

transition amplitude $\langle q_f, t_f | q_i, t_i \rangle = \int Dq(t) \exp(iS[q]/\hbar)$ integrated over all paths $q(t)$ connecting (q_i, t_i) to (q_f, t_f) . Wick rotation $t \rightarrow -i\tau$ converts oscillatory integral to convergent Gaussian for non-perturbative computation. Quantum mechanics + QFT path integrals reformulate Schrödinger + Heisenberg pictures. BST substrate cross-link: per Vol 5 Ch 5 + Vol 1 Ch 7 + Ch 9, substrate-tick $GF(2^g) = GF(128)$ per Koons tick (T2429 Reed-Solomon discretization + K59 7-step cyclotomic mechanism RATIFIED); each Koons tick contributes one finite-step propagation operator U_{tick} ; integrated over many ticks gives continuum path integral as finite sum (NOT infinite-dimensional integral). Per T2457 (Friday Cal #92(b) compliant framing): Bergman reproducing kernel $K_B(z, \bar{w}) = c_{\text{FK}} \cdot h(z, \bar{w})^{-(g/\text{rank})}$ plays structural role of QFT Feynman propagator with three structural advantages: positive-definite (Bergman 1922) $\rightarrow$ no $i\epsilon$ prescription needed; UV-complete (substrate-tick UV cutoff at $N_{\text{max}} = 137$ per T2437) $\rightarrow$ no UV divergences; BST primary normalization $c_{\text{FK}} = 225/\pi^{(9/2)}$ (T2442 EXACT). No Wick rotation required because substrate-tick $GF(128)^k$ is real-time finite-step (T2405 Koons tick = $t_{\text{Planck}} \cdot \alpha^{\{C_2^2\}} \approx 10^{-120}$ s). Standard QFT Feynman diagrams reduce to substrate-tick computation chains; α -vertex insertions per T2476 substrate-coordinate-count $k(P)$ substrate-mechanism (Friday); cross-link Vol 5 Ch 6 hydrogen atomic corrections.

Level 3 (5th-grader accessible): Calculus of variations finds extremum (max/min) of integrals (like the shortest path between two points). Hamilton's principle of stationary action makes classical mechanics elegant. Feynman extended this to quantum mechanics with path integrals — sum over ALL possible paths a particle could take, weighted by $\exp(iS/\hbar)$. BST shows the substrate's $GF(128)$ coding (Vol 5 Ch 5 + T2429) makes this an explicit FINITE sum at substrate-tick scale.

Section 10.1 — Standard Variational Principles

Stationary action $\delta S = 0$ with $S = \int L dt$; Euler-Lagrange $\partial L / \partial q - d/dt (\partial L / \partial \dot{q}) = 0$.

Lagrangian $L = T - V$; Hamiltonian $H = p\dot{q} - L$; canonical equations $dq/dt = \partial H / \partial p$, $dp/dt = -\partial H / \partial q$.

Noether's theorem: continuous symmetry $\rightarrow$ conserved charge.

Section 10.2 — Feynman Path Integral (1948)

$\langle q_f, t_f | q_i, t_i \rangle = \int Dq(t) \exp(iS[q]/\hbar)$ sum over all paths.

Wick rotation $t \rightarrow -i\tau$ converts to convergent Gaussian.

QM + QFT path integrals reformulate Schrödinger + Heisenberg pictures.

Section 10.3 — Substrate Path Integral via $GF(128)^k$ (Vol 5 Ch 5 cross-link)

Per Vol 5 Ch 5 + Vol 1 Ch 7 + Ch 9: substrate-tick $GF(2^g) = GF(128)$ per Koons tick (T2429 + K59 RATIFIED).

Each Koons tick contributes one finite-step propagation operator U_{tick} :

$$|\psi(t_f)\rangle = U_{\text{tick}}^N |\psi(t_i)\rangle \text{ where } N = (t_f - t_i) / \tau_{\text{Koons}}$$

Continuum path integral = integrated-state limit over many substrate ticks. FINITE sum (NOT infinite-dimensional integral).

Section 10.4 — T2457 Bergman = Feynman Propagator (Cal #92(b))

T2457 (Friday): Bergman reproducing kernel $K_B(z, \bar{w}) = c_{\text{FK}} \cdot h(z, \bar{w})^{-(g/\text{rank})}$ plays structural role of QFT Feynman propagator.

Three structural advantages: 1. Positive-definite (Bergman 1922) $\rightarrow$ no $i\epsilon$ prescription 2. UV-complete ($N_{\text{max}} = 137$ cutoff per T2437) $\rightarrow$ no UV divergences 3. BST primary normalization $c_{\text{FK}} = 225/\pi^{(9/2)}$ (T2442 EXACT)

No Wick rotation needed because substrate-tick $GF(128)^k$ is real-time finite-step.

Section 10.5 — Honest scope + Connection

- Standard calculus of variations + Feynman path integral [?]
- Substrate path integral via T2429 $GF(128)^k$ + T2457 Bergman propagator [?]
- T2476 $\alpha^{\{\text{BST primary}\}}$ exponent pattern cross-link (per-process substrate-coordinate count)
- Open scope: explicit Feynman diagram $\rightarrow$ substrate-tick chain reduction (multi-week)

Connection: - Vol 5 Ch 5 Heisenberg + Path Integral - Vol 1 Ch 7 Dynamics + Ch 9 Scattering - T2457 Bergman propagator (Cal #92(b)) - T2476 $\alpha^{\{\text{BST primary}\}}$ exponent pattern (Friday cross-CI peak)

— Lyra, Vol 10 Ch 10 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT